

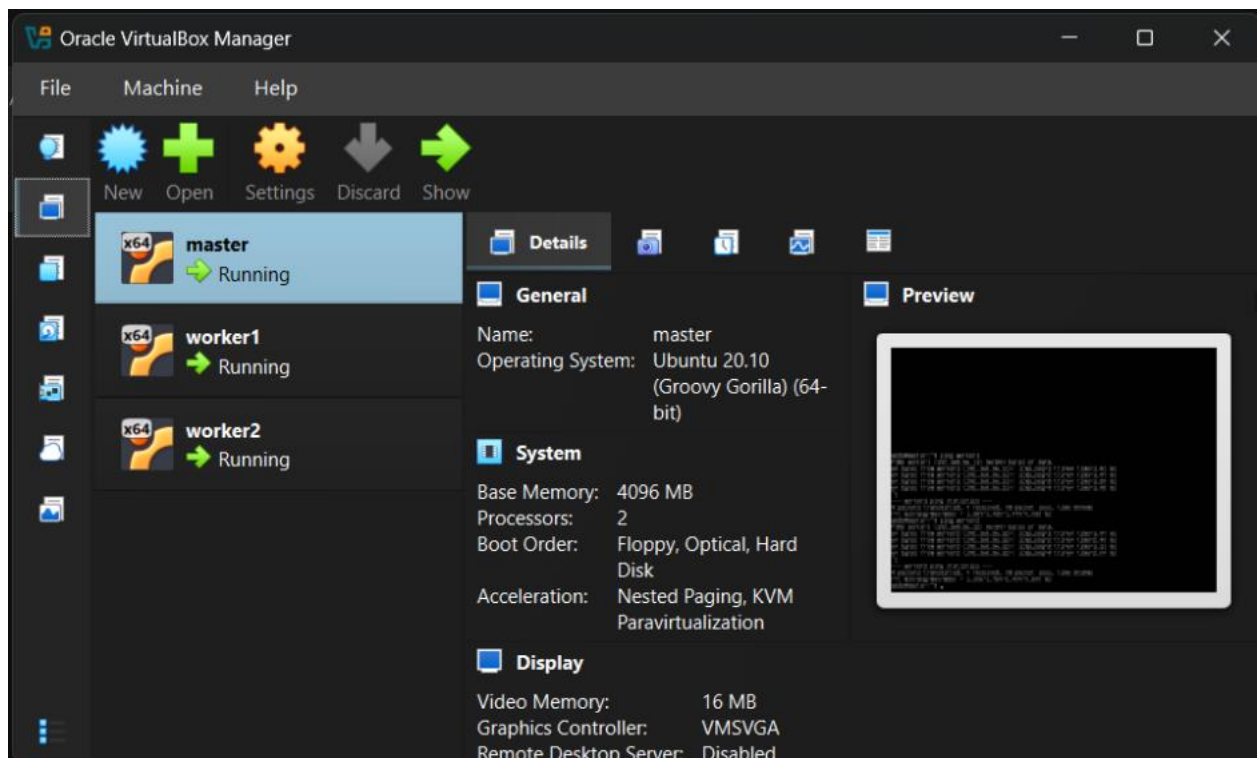


Figure 1: VirtualBox showing the three created VMs: master, worker1, and worker2.

```
master [Running] - Oracle VirtualBox
File Machine View Input Devices Help

    valid_lft 86253sec preferred_lft 14253sec
    inet6 fe80::a00:27ff:fec8:a30d/64 scope link
    valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:99:68:e5 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.10/24 scope global enp0s8
    valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe99:68e5/64 scope link
    valid_lft forever preferred_lft forever
abdo@master:~$ hostname
master
abdo@master:~$ sudo ip addr del 192.168.56.10/24 dev enp0s8
abdo@master:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
    valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c8:a3:0d brd ff:ff:ff:ff:ff:ff
    inet6 fd17:625c:f037:2:a00:27ff:fec8:a30d/64 scope global dynamic mngtmpaddr noprefixroute
    valid_lft 86145sec preferred_lft 14145sec
    inet6 fe80::a00:27ff:fec8:a30d/64 scope link
    valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:99:68:e5 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.10/24 scope global enp0s8
    valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe99:68e5/64 scope link
    valid_lft forever preferred_lft forever
abdo@master:~$ hostname
master
abdo@master:~$ _
```

Figure 2: Output of ip a on the master node showing 192.168.56.10/24.

```
worker2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

abdo@worker2:~$ ping master
PING master (192.168.56.10) 56(84) bytes of data:
64 bytes from master (192.168.56.10): icmp_seq=1 ttl=64 time=1.69 ms
64 bytes from master (192.168.56.10): icmp_seq=2 ttl=64 time=1.32 ms
64 bytes from master (192.168.56.10): icmp_seq=3 ttl=64 time=2.00 ms
64 bytes from master (192.168.56.10): icmp_seq=4 ttl=64 time=1.10 ms
^C
--- master ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.100/1.478/2.002/0.385 ms
abdo@worker2:~$ ping worker1
PING worker1 (192.168.56.11) 56(84) bytes of data:
64 bytes from worker1 (192.168.56.11): icmp_seq=1 ttl=64 time=1.57 ms
64 bytes from worker1 (192.168.56.11): icmp_seq=2 ttl=64 time=1.70 ms
64 bytes from worker1 (192.168.56.11): icmp_seq=3 ttl=64 time=1.89 ms
64 bytes from worker1 (192.168.56.11): icmp_seq=4 ttl=64 time=1.60 ms
64 bytes from worker1 (192.168.56.11): icmp_seq=5 ttl=64 time=1.01 ms
^C
--- worker1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4012ms
rtt min/avg/max/mdev = 1.006/1.541/1.825/0.282 ms
abdo@worker2:~$ _

worker1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

abdo@worker1:~$ ping master
PING master (192.168.56.10) 56(84) bytes of data:
64 bytes from master (192.168.56.10): icmp_seq=1 ttl=64 time=1.97 ms
64 bytes from master (192.168.56.10): icmp_seq=2 ttl=64 time=1.64 ms
64 bytes from master (192.168.56.10): icmp_seq=3 ttl=64 time=2.18 ms
64 bytes from master (192.168.56.10): icmp_seq=4 ttl=64 time=1.64 ms
^C
--- master ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3032ms
rtt min/avg/max/mdev = 1.642/1.858/2.179/0.227 ms
abdo@worker1:~$ ping worker2
PING worker2 (192.168.56.12) 56(84) bytes of data:
64 bytes from worker2 (192.168.56.12): icmp_seq=1 ttl=64 time=2.50 ms
64 bytes from worker2 (192.168.56.12): icmp_seq=2 ttl=64 time=2.52 ms
64 bytes from worker2 (192.168.56.12): icmp_seq=3 ttl=64 time=1.34 ms
64 bytes from worker2 (192.168.56.12): icmp_seq=4 ttl=64 time=2.02 ms
^C
--- worker2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 1.341/2.095/2.523/0.478 ms
abdo@worker1:~$ _
```

Figure 3: Ping test from master to worker1 & worker2 showing 0% packet loss.

```

abdo@master:~$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/abdo/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/abdo/.ssh/id_rsa
Your public key has been saved in /home/abdo/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:M0I4sMzHL13p0utVKUIk8NgZfsDwqKafLopfgMA32i8 abdo@master
The key's randomart image is:
+---[RSA 3072]-----+
|  . 0+++.0 |
| .o + 0.B . |
| ..+0B X o o |
| ..+=.* . . . |
| ..=,+ + S . |
| o o.. o + |
| . E... . |
| o. o.. . |
| ++= . |
+-----[SHA256]-----+
abdo@master:~$ _

```

```

| . E... . |
| o. o.. . |
| ++= . |
+-----[SHA256]-----+
abdo@master:~$ ssh-copy-id abdo@worker1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/abdo/.ssh/id_rsa.pub"
The authenticity of host 'worker1 (192.168.56.11)' can't be established.
ECDSA key fingerprint is SHA256:21ALncqyjMB96pYxsAEpLxy7BBCvSPv79WJKDIgcFvA.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install all the new keys
abdo@worker1's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'abdo@worker1'"
and check to make sure that only the key(s) you wanted were added.

abdo@master:~$ ssh-copy-id abdo@worker2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/abdo/.ssh/id_rsa.pub"
The authenticity of host 'worker2 (192.168.56.12)' can't be established.
ECDSA key fingerprint is SHA256:Ao4wg9MeMLg3GdL1wFXiPnKJGKMBtOK+Kr3BQhLQHpy.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install all the new keys
abdo@worker2's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'abdo@worker2'"
and check to make sure that only the key(s) you wanted were added.

abdo@master:~$

```

```
master [Running] - Oracle VirtualBox
Usage of /:                27.2% of 9.75GB
Memory usage:              5%
Swap usage:                0%
Processes:                 117
Users logged in:           1
IPv4 address for enp0s3:   10.0.2.15
IPv6 address for enp0s3:   fd17:625c:f037:2:a00:27ff:fe99:869a
IPv4 address for enp0s8:   192.168.56.11

* Ubuntu 20.04 LTS Focal Fossa has reached its end of standard support
  on 31 May 2025.

For more details see:
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0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Mon May 25 17:05:09 2026 from 192.168.56.10
abdoo@worker1:~$ _

master [Running] - Oracle VirtualBox
Swap usage:                0%
Processes:                 124
Users logged in:           1
IPv4 address for enp0s3:   10.0.2.15
IPv6 address for enp0s3:   fd17:625c:f037:2:a00:27ff:fed1:a7d
IPv4 address for enp0s8:   192.168.56.12

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For more details see:
https://ubuntu.com/20-04

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0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Mon May 25 15:32:54 2026
abdoo@worker2:~$ hostname
worker2
abdoo@worker2:~$ _
```

Figure 4: Passwordless SSH login from master to worker1 and worker2, showing successful hostname verification.

```

abdo@master:~$ mpirun -np 6 --hostfile hostfile --mca btl_tcp_if_include enp0s8 --mca oob_tcp_if_include enp0s8 --mca btl ^openib --wdir /tmp/hpc python3 distributed_mnist.py
Scores from all nodes: [1.0, 1.0, 1.0, 1.0, 1.0, 1.0]
Average score: 1.0
abdo@master:~$ |

```

Figure 5: Output of the distributed machine learning job showing six node scores and the average score.

```

abdo@master:/tmp/hpc$ mpirun -np 6 --hostfile ~/hostfile --mca btl_tcp_if_include enp0s8 --mca oob_tcp_if_include enp0s8 --mca btl ^openib --wdir /tmp/hpc python3 distributed_gene_analysis.py
Bioinformatics scores from all nodes: [1.0, 1.0, 1.0, 1.0, 1.0, 1.0]
Average score: 1.0

```

Figure 6: Output of the distributed bioinformatics job showing six scores and the average score.

```

abdo@master:~$ sudo docker node ls
[sudo] password for abdo:
ID                HOSTNAME    STATUS    AVAILABILITY    MANAGER ST
ATUS  ENGINE VERSION
ku7ct91esh8dzf1c4uoehimv3 *  master     Ready     Active          Leader
26.1.3
owjuv471aord4whpxqn66wx06    worker1     Ready     Active
26.1.3
o5e0v35mw9w4wzjzoanwg24sj    worker2     Ready     Active
26.1.3
abdo@master:~$

```

Figure 7: Docker Swarm node list showing master, worker1, and worker2 in Ready state.

```

abdo@master:~$ sudo docker service ls
ID                NAME                MODE                REPLICAS    IMAGE                PORT
S
rxihtukcijhz      hpc_spark-master    replicated          1/1          spark:3.5.0         *:70
77->7077/tcp, *:8080->8080/tcp
cu8uj045424o      hpc_spark-worker    replicated          2/2          spark:3.5.0
abdo@master:~$ sudo docker stack ps hpc
ID                NAME                IMAGE                NODE          DESIRED STATE
CURRENT STATE    ERROR              PORTS
0s1laaffx17yg     hpc_spark-master.1  spark:3.5.0         master       Running
Running 5 minutes ago
miwex7mdje7j      hpc_spark-worker.1  spark:3.5.0         master       Running
Running 5 minutes ago
lfhadwysadoy      hpc_spark-worker.2  spark:3.5.0         master       Running
Running 5 minutes ago
abdo@master:~$ |

```

Figure 8: Docker service list showing Spark master 1/1 and Spark worker 2/2 and Spark services running.

```
26/05/27 17:05:57 INFO DAGScheduler: Job 2 finished: count at /tmp/distribut
ed_gene_expression_analysis.py:61, took 1.382539 s
Distributed Spark ML completed successfully
Bioinformatics dataset rows: 200
Number of gene features: 20
Test accuracy: 1.0
```

Figure 9: PySpark job output showing successful execution and test accuracy of 1.0.